Finding Barium Dwarfs Among Established White-Dwarf Systems

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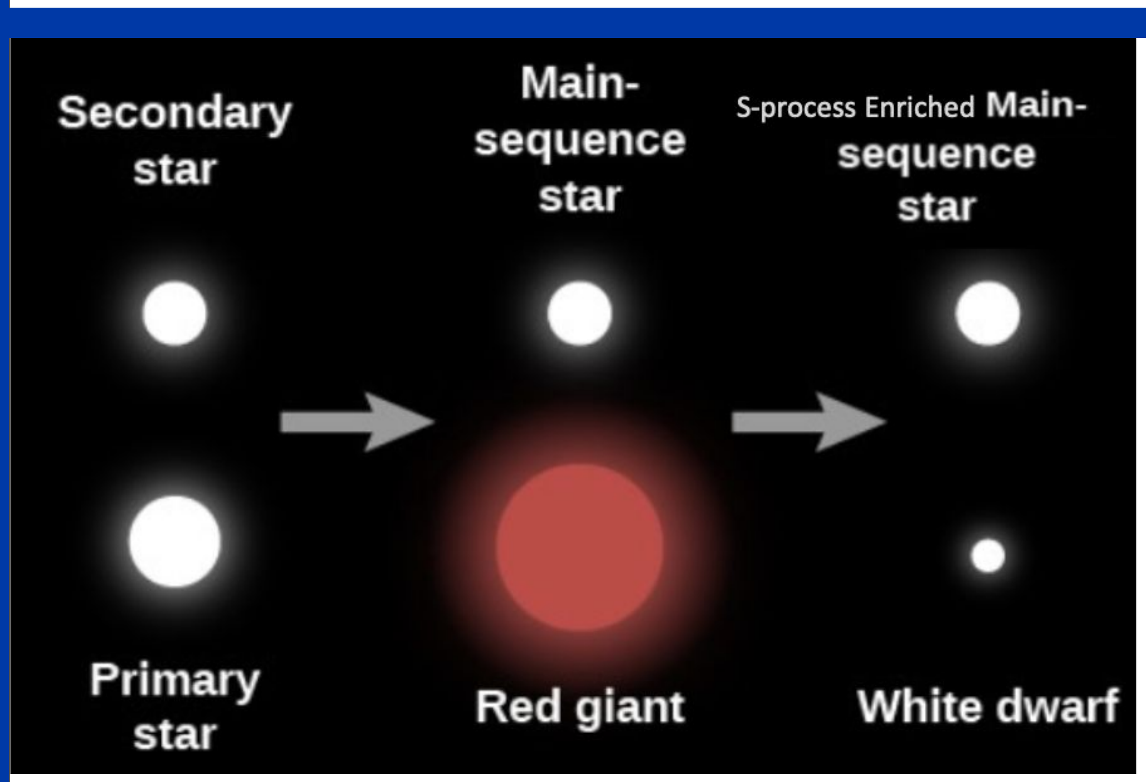
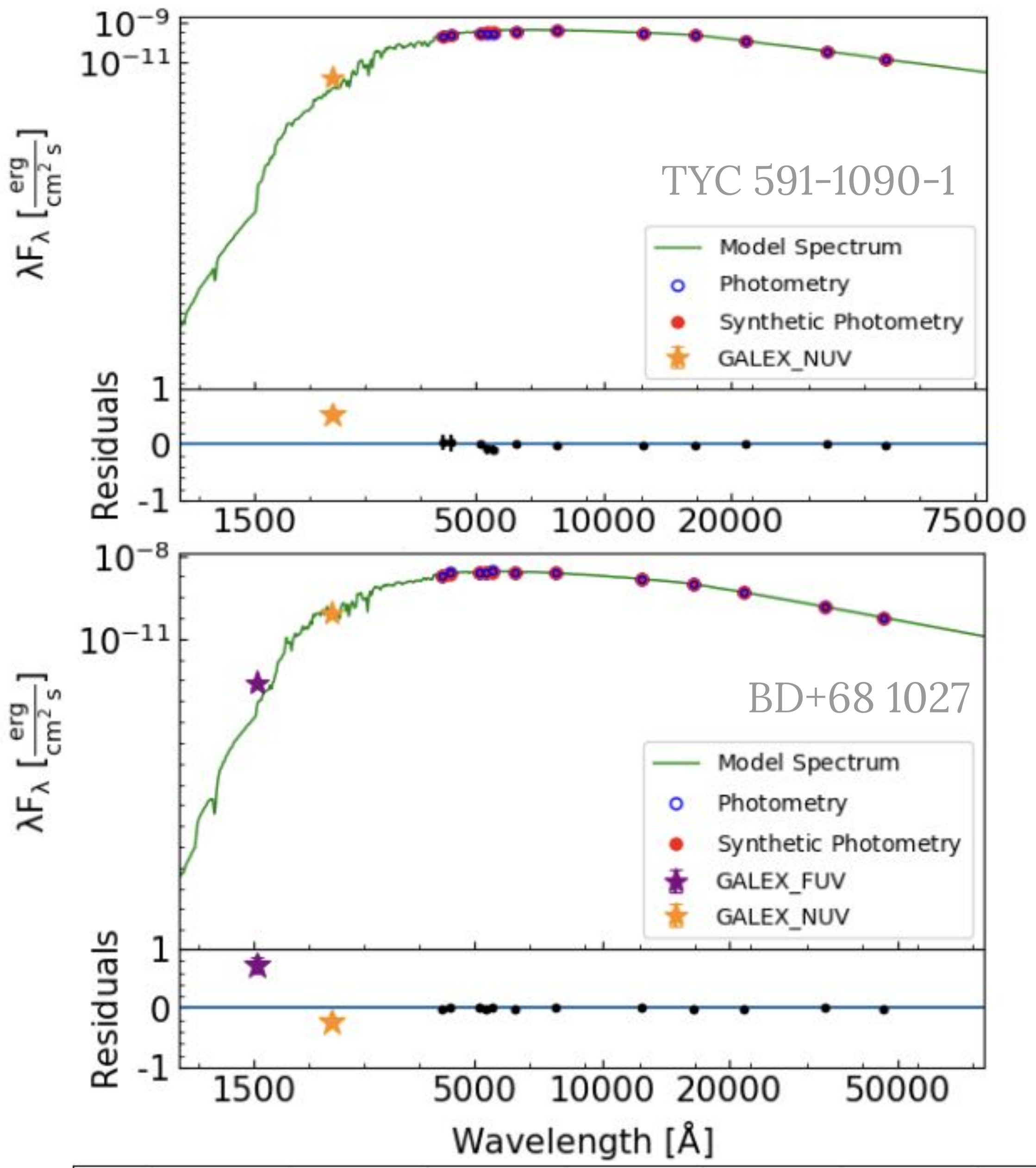
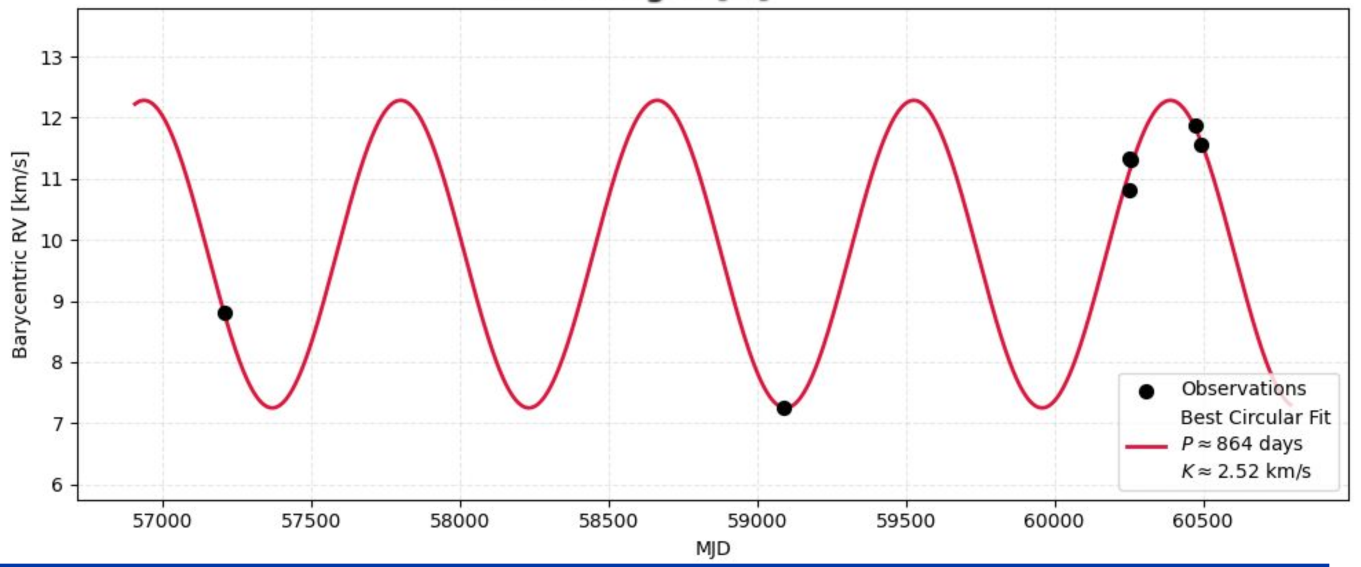


Two Example Ba. Dwarfs





Cosmic Pollution & Binary Evolution

Barium dwarfs (Ba. dwarfs) are main-sequence stars enriched by past mass transfer. They host a "hidden" companion: a white-dwarf (WD) that was once a giant star. During its final stages, this giant produced heavy s-process elements (Ba, Sr, La) and transferred this enriched material onto its companion. This study aims to identify these "polluted" systems within known white-dwarf binaries to better understand how stars exchange mass.

The Hunt: Three Ways to Find a Hidden Star

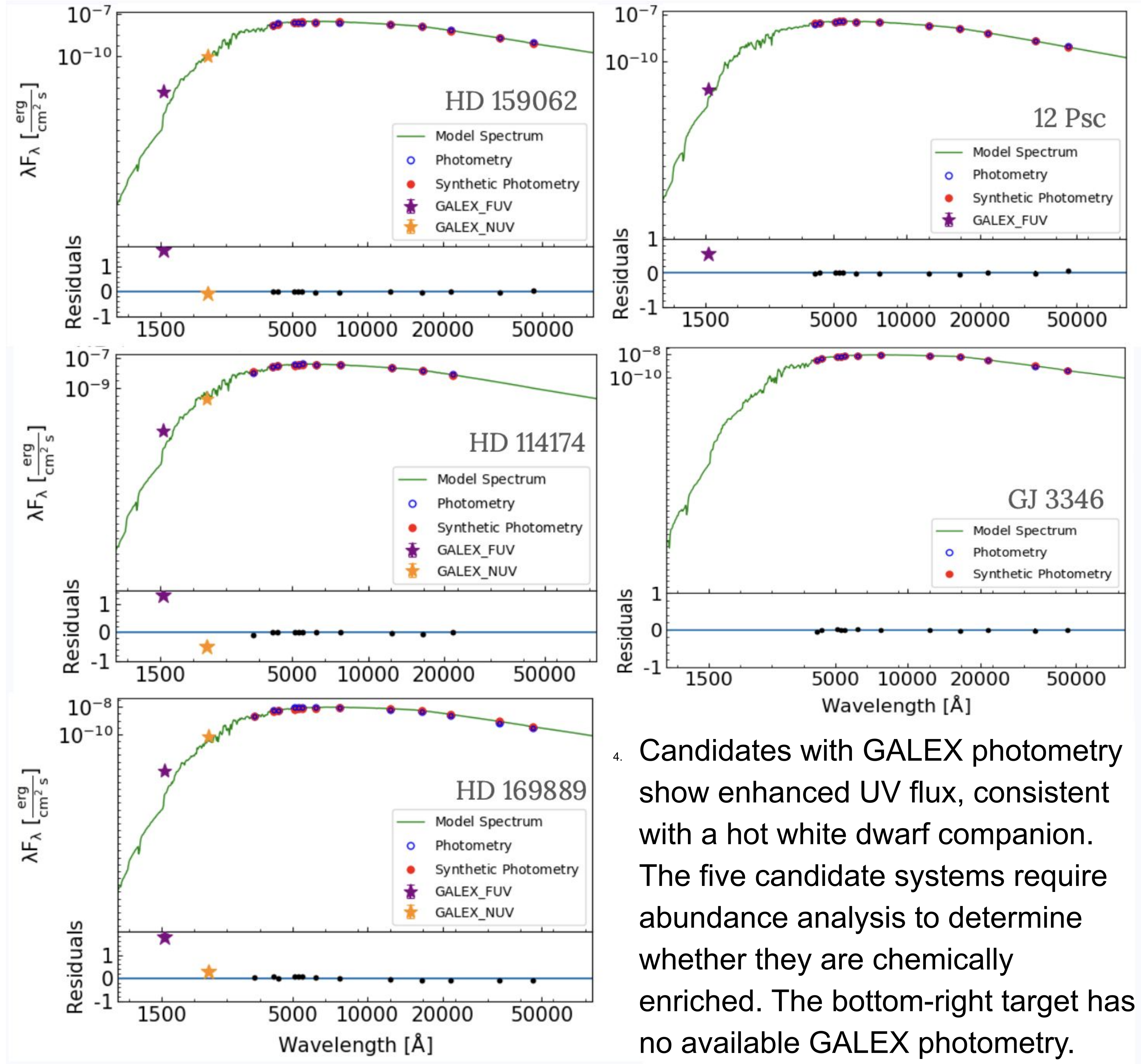
Because the white-dwarf companion is often too faint to detect in visible light, we use three diagnostic tools to test for its presence:

UV Excess (SED Fitting): We construct photosphere-only spectral energy distributions (SEDs) using RADPy SEDFit. While the primary star dominates the visible light, a hot white-dwarf will produce a measurable UV excess in GALEX NUV/FUV data relative to the photospheric fit

Radial Velocity (The "Wobble"): We are compiling and analyzing historical radial-velocity measurements from archival spectra to trace long-term orbital motion. A changing radial velocity indicates that the primary star is being pulled by an unseen companion. The example shown is TYC 591-1090-1 over a 9-year baseline.

Chemical Fingerprinting (In Progress): We are using MOOG stellar atmosphere modeling to perform abundance analysis. By comparing observed spectra to synthetic models, we estimate the level of Barium enrichment and test for the chemical signature of past mass transfer.

Candidate Ba. Dwarfs



Candidates with GALEX photometry show enhanced UV flux, consistent with a hot white dwarf companion. The five candidate systems require abundance analysis to determine whether they are chemically enriched. The bottom-right target has no available GALEX photometry.

Star Name	FUV σ	NUV σ	P	K	[Fe/H]	[Ba/Fe]	Source
TYC 591-1090-1	+5.68	-74.30	864 d	2.52 km/s	0.00	Ongoing	McGuire (2026)
BD+68 1027	N/A	+26.66	425 d	2.34 km/s	-0.44	+1.11	Escorza (2019)
HD 159062	+22.92	-41.04	~411 yr	Linear Trend	-0.38	+0.40	Fuhrmann (2017)
HD 114174	+18.73	-376.53	12.2 yr	Acceleration	+0.05	+0.1 to +0.3	Crepp (2013)
HD 169889	+11.83	+70.05	2.25 yr	1.7 km/s	+0.01	In Prep	Venner (2023)
12 Psc	+42.00	N/A	50–100 yr	Linear Trend	+0.01	In Prep	Bowler (2021)
GJ 3346	N/A	N/A	Long-term	Acceleration	+0.05	In Prep	Bonavita (2020)

ACKNOWLEDGEMENTS

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Fuhrmann, K., et al. 2017, *ApJ*, 836, 139.
Gaia Collaboration, et al. 2023, *A&A*, 674, A1.
Roriz, M., et al. 2024, *MNRAS*, 527, 11072.
Sneden, C. 1973, *ApJ*, 184, 839. (MOOG)
Venner, A., et al. 2023, *MNRAS*, 524, 695.

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